

Joshua C. Laferriere, M.S.

Full Stack AI/ML Engineer

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thistleknot.github.io/professional_portfolio | github.com/thistleknot

SUMMARY

12+ years building production ML and data systems across aerospace, media, and government infrastructure. Specializes in NLP, semantic retrieval, and recommendation systems. Ships production code; integrates current research into working systems.

CORE COMPETENCIES

Transformers & Vision Transformers | Deep Learning & NLP | RAG & Semantic Retrieval | Reinforcement Learning | Time Series | Bayesian Methods

PyTorch · TensorFlow · LangGraph · LangChain · spaCy · Streamlit · Plotly · FastMCP · FastAPI · React · Flask

PostgreSQL · pgVector · Neo4j · Dask · Airflow · SQL | Python · R · Bash · C++

AWS · GCP · Azure · Docker · Kubernetes · CloudFoundry · GitLab CI/CD | NIST/DFARS Compliance

Storage · GlusterFS · LustreFS · VDO | **Presentation** · Grafana · Plotly

PROFESSIONAL EXPERIENCE

Data Scientist & AI Engineer — Boeing

Seal Beach, CA (Hybrid) | Feb 2022 – Present

- **RAG pipeline** → surfaced relevant documentation from unstructured corpus → *reduced service cycle time for maintenance operations*
- **Full stack RAG deployment** → built end-to-end semantic retrieval system with React front end and FastAPI back end for safety reporting → *reduced reporting time from hours to minutes*
- **Semantic modeling** → mapped component lifecycle relationships into a queryable business layer → *structured operational support across maintenance workflows*
- **NLP pipeline modernization** → replaced legacy entity recognition with current architecture → *eliminated end-of-life dependency risk; domain accuracy maintained*
- **Distributed data processing** → introduced Dask for scalable pipeline execution → *scaled data workflows beyond single-machine capacity*
- **Container-based deployment** → operationalized Docker and CloudFoundry for ML application hosting → *reproducible, portable delivery under compliance requirements*
- **Quality-gated CI/CD pipeline** → authored unit and regression test suites as automated pipeline blockers; configured GitLab runner for continuous delivery; manual QA reserved for release sign-off → *end-to-end ownership of build, test, and deploy chain across data science and engineering*
- **Distributed storage** → deployed GlusterFS and VDO for shared compute storage with deduplication → *optimized storage utilization across simulation and processing environments*
- **Cross-functional coordination** → aligned data science and engineering delivery via Jira and Azure DevOps → *consistent program output across both teams*

AI Engineer Consultant — Vennify.com

Remote | Nov 2023 – Feb 2024

- **Controlled deployment methodology** → validated model rollout via structured testing → *15% inference latency reduction*
- **Training pipeline optimization** → restructured data batching strategy → *30% reduction in training time*

Systems Engineer III — Boeing (Contractor)

Oklahoma City, OK | Aug 2021 – Feb 2022

- **Identity management** → unified authentication across heterogeneous OS environments → *DFARS/NIST compliance across distributed infrastructure*

- **Infrastructure monitoring** → deployed visibility tooling across distributed systems → *operational awareness for engineering and support teams*
- **Lifecycle tracking** → built software end-of-life database → *structured migration planning and execution support*

System Administrator II — NOAA (Contractor)

Norman, OK (Hybrid) | Jun 2020 – Aug 2021

- **HPC cluster operations** → maintained high-performance computing environment → *uninterrupted scientific computing for weather forecasting*

Linux Systems Engineer — Quadranet

Los Angeles, CA | Oct 2018 – Dec 2018

- **Cloud infrastructure** → provisioned and managed hosted client environments

Linux Systems Engineer / QA Support — DirecTV

El Segundo, CA | Jan 2015 – Mar 2019

- **Deployment automation and QA validation** → scripted firmware rollout with embedded verification across network equipment → *eliminated manual deployment bottleneck; quality checks integral to release process*
- **Linux rack support** → deployed resilient server systems interfacing with network devices → *stable physical infrastructure supporting QA and delivery operations*

PROJECTS & ACCOMPLISHMENTS

- **Music discovery system** → clustered songs into navigable genre and artist groupings → *listener can explore by similarity or pivot intentionally across genres*
- **Signed similarity graph** → positive and negative traversal paths from a single mechanism → *surfaces what is near and what is deliberately far*
- **Artist-to-genre relationship mapping** → quantifies whether an artist defines a genre or merely appears in it → *explainable, user-navigable recommendations*
- **Language model fine-tuning** → adapted model to domain-specific reasoning tasks → *production-ready inference with reward-shaped behavior*
- **Agentic orchestration** → coordinated multi-agent workflows across tool-using systems → *autonomous task decomposition at scale*
- **Investment research pipeline** → systematic data mart for portfolio fundamentals → *quantitative decision-making over market and macro data*
- **Automated literature synthesis** → research retrieval pipeline over academic corpus → *continuous integration of current research into active projects*
- **Labor optimization algorithm** → contributed workforce management feature to OSS project → *deployed in production tool suite; demonstrated at developer conference*
- **Reinforcement learning game engine** → trained progressive game-playing agent → *value-based learning combined with strategic search*

Portfolio: thistleknot.github.io/professional_portfolio

EDUCATION

Master of Science in Information Technology (Data Science)

California State University Fullerton | Aug 2017 – May 2019

Bachelor of Science in Business Administration (Business Analytics)

California State University Dominguez Hills | Aug 2013 – Dec 2016

Associate of Science in Computer Programming

Long Beach City College | Aug 2013 – Dec 2016